Table 10.3: Functions of proteins			
1	Build and repair body tissues		Proteins form integral parts of most body structure such as skin, tendon, membranes, muscles, organs and bones. They support the growth and repair of body tissues.
2	**Regulation of body processes:** Proteins are required for highly specialized functions in our body. These proteins are as follows:		
	a.	Immune proteins	Antibodies, necessary for immunity reactions, are protein in nature. Resistance to disease is an immunological response.
	b.	Hormones	Regulates body processes. Hormones such as adrenocorticotrophic hormone (ACTH) and insulin, are protein in nature
	c.	Enzymes	All enzymes are protein in nature and are required at every step of digestion, absorption, and metabolism
	d.	Nucleoproteins	These govern the synthesis of all body proteins (e.g) Histones, protamine
	e.	Contractile proteins	Actin and myosin are responsible for the action of muscles.
	f.	Blood proteins	Haemoglobin is a protein which carries oxygen. Other proteins found in blood are lipoproteins, transferrin, serum albumin and immunoglobulins. Serum albumin is also responsible for regulating osmotic pressure and maintaining the fluid balance of the body
	g.	Specific functions	Some amino acids have specific and specialized functions in the body • Tryptophan is a precursor of niacin and serotonin. Methionine supplies labile methyl groups for synthesizing choline, which prevents accumulation of fat in the liver • Glycine is required for the formation of the porphyrin ring of haemoglobin and is an important constituent of nucleic acids
3	Supply of energy		Proteins provide fuel for the body's energy needs [4 Kcal/gm].
4	Storage		Proteins help to store iron and copper.
5	Acid-base balance		Proteins help maintain the acid-base balance of the body fluids by acting as buffers.
6	Pregnant and lactating women		Provides amino acids for the growth of foetus in pregnancy and for the production of milk during lactation.

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